LITERATURE REVIEW

Sinkhorn Distances: Lightspeed Computation of Optimal Transportation Distances

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ANALYTICAL ABSTRACT

The paper asks whether the optimal transportation (earth mover's) distance between histograms, whose exact computation costs at least cubic time in the dimension, can be replaced by a smoothed version that is cheap to compute and at least as useful for learning. Cuturi restricts the feasible transport plans to a Kullback-Leibler ball around the independence table, or equivalently adds an entropy penalty weighted by a multiplier, and observes that the penalised optimum has the form of a diagonally scaled Gibbs kernel. The scaling vectors are then obtained by the Sinkhorn-Knopp matrix scaling iteration, which consists of matrix products and parallelises on graphics processors. The paper proves that the hard-constrained version is symmetric and obeys the triangle inequality, that it recovers the classical distance for a loose constraint, and that it becomes a negative definite kernel in the independent limit. Experiments on MNIST digits show lower classification error than the exact distance and several other histogram distances, and speed gains above five orders of magnitude. The review reconstructs the derivations, shows that the metric guarantees do not transfer to the computed quantity, and reports a small independent numerical study of its bias, convergence and non-coincidence.

1 Introduction

1.1 Paper identification and classification

The paper is a methodological contribution to computational statistics and machine learning with a theoretical component (a metric result and two kernel results) and a limited empirical component (one classification benchmark and two timing studies). Its field is optimal transport applied to histogram comparison, and its subfield is regularised transport and matrix scaling. The version read in full is the arXiv preprint 1306.0895, in its HTML rendering, which contains the complete text, all equations, the algorithm listing, the appendix proofs and the reference list. The figures could not be inspected as images in this rendering, so Figures 2 to 5 of the paper are analysed through their captions and the surrounding text only, and any statement about their shape is labelled accordingly.

1.2 Executive understanding

WHAT IS THE PAPER TRYING TO DISCOVER?

Whether a transport distance can be made computable at the scale of modern data (histograms with hundreds or thousands of bins) without restricting the ground metric, and whether doing so costs accuracy. The discovery claimed is that the cost can be removed by entropic smoothing and that accuracy even improves.

WHY DOES THE QUESTION MATTER?

Transport distances are the only common histogram distances that are parameterised by a ground metric encoding which bins are similar. This is what allows them to handle redundant or neighbouring features, as a Mahalanobis distance does for correlated coordinates. If they cannot be computed, that advantage is unusable in large-scale learning, retrieval and, as the later literature shows, in generative modelling, barycentre computation and distributionally robust optimisation.

WHAT WAS KNOWN BEFORE, AND WHAT IS MISSING?

Known: the optimum of the transport linear programme is a distance when the cost matrix is a metric (Villani, 2009), its cost is at least of order $d^3 \log d$ for general ground metrics (Pele and Werman, 2009), and fast variants exist only under restrictions on the ground metric or with approximation (Grauman and Darrell, 2004). Entropy-regularised transport was long used in transportation planning (Erlander and Stewart, 1990). Missing: a regularised transport quantity connected to a fast, vectorisable algorithm, with a statement of which metric properties survive, and evidence that it works in learning.

WHAT DOES THE PAPER PROPOSE, HOW DOES IT INVESTIGATE, AND WHAT DOES IT FIND?

It proposes the Sinkhorn distance $d_{M,\alpha}$ (entropy-constrained plans) and the dual-Sinkhorn divergence d_M^λ (entropy-penalised plans), computed by Algorithm 1. It investigates them with short proofs and three experiments. It finds that d_M^λ beats the exact distance and four other distances in support vector machine classification of MNIST digits, that it stays about ten percent above the exact distance even for large λ, and that it is more than 10^5 times faster on one CPU core for large dimensions.

2 Research Problem and Hypothesis

2.1 Research question

PAPER The paper asks whether adding an entropic constraint or penalty on the transport plan between two histograms r and c yields a quantity that remains a useful distance, can be computed by matrix scaling orders of magnitude faster than a transport solver for an arbitrary ground metric M, and gives equal or better classification performance than the exact transport distance.

2.2 Hypotheses

- H1 (metric): for every $\alpha \geq 0$ and every metric cost matrix M, the entropy-constrained optimum $d_{M,\alpha}$ is symmetric and satisfies the triangle inequality.
- H2 (limits): for large α it equals the classical transport distance, and for $\alpha = 0$ it equals $r^T M c$, a negative definite kernel when M is a Euclidean distance matrix.
- H3 (computation): the penalised version is solved by Sinkhorn-Knopp scaling with linear convergence, vectorised over many target histograms and suited to graphics processors.
- H4 (performance): the smoothed distance performs at least as well as the exact one in learning, because smoother plans are a more robust notion of dissimilarity.

2.3 Existing literature and research gap

PAPER The authors position the work against exact solvers (network simplex, interior point), restricted-metric or approximate solvers, and against the regularised transport of Ferradans et al. (2013), who penalise plan irregularity through graph norms for colour transfer. The gap is a regularisation that is statistically motivated (maximum entropy; Jaynes, 1957) and algorithmically exploitable.

CRITIQUE The gap is real for computation. For the statistical motivation, the argument that smooth plans are ”more plausible” is asserted through a schematic figure and an appeal to the maximum-entropy principle, not tested against an alternative regulariser in the experiments, so the paper cannot separate the benefit of entropy from the benefit of any smoothing.

2.4 Claimed contribution

CRITIQUE Three contributions are claimed: a family of distances, a fast algorithm, and empirical superiority. The algorithmic contribution is demonstrated convincingly for the dual quantity. The metric contribution is demonstrated for $d_{M,\alpha}$, which is not what is computed. The empirical superiority is demonstrated on one dataset with tuned hyperparameters per method, which supports a conditional claim only (Section 6).

3 Theoretical Framework and Mechanism

3.1 Transport polytope, entropy and the entropic ball

PAPER For histograms r, c in the simplex $\Sigma_d = \{x \in \mathbb{R}_+^d : x^T \mathbf{1}_d = 1\}$, the transport polytope is the set of nonnegative matrices with prescribed marginals, which is the set of joint probabilities of two discrete variables $X \sim r$ and $Y \sim c$.

$$U(r,c) = \{P \in \mathbb{R}_+^{d\times d} : P\mathbf{1}_d = r,\ P^T\mathbf{1}_d = c\}, \qquad d_M(r,c) = \min_{P\in U(r,c)} \langle P, M\rangle. \tag{1}$$

P	Transport plan; p_{ij} is the mass moved from bin i to bin j, or $\Pr(X=i, Y=j)$.
M	Cost matrix. It is a distance matrix when $m_{ii} = 0$ and $m_{ij} \le m_{ik} + m_{kj}$.
$\langle P, M\rangle$	Frobenius product $\sum_{ij} p_{ij} m_{ij}$, the total cost of the plan.

Equation (1) is a linear programme whose optimum sits at a vertex of $U(r,c)$, a matrix with at most $2d-1$ nonzero entries. Such a plan is nearly deterministic: each source bin sends its mass to very few destinations. The paper's idea is to move away from vertices. With $h(P) = -\sum_{ij} p_{ij} \log p_{ij}$ and $h(r) = -\sum_i r_i \log r_i$, every plan satisfies

$$h(P) \le h(r) + h(c), \qquad \mathrm{KL}(P \,\|\, rc^T) = h(r) + h(c) - h(P) = I(X;Y), \tag{2}$$

with equality only at the independence table rc^T. The identity follows by expanding $\sum_{ij} p_{ij} \log \frac{p_{ij}}{r_i c_j} = -h(P) - \sum_{ij} p_{ij} \log r_i - \sum_{ij} p_{ij} \log c_j$ and using the marginals to get $h(r) + h(c)$ in the last two terms. The Sinkhorn distance restricts the feasible set to a Kullback-Leibler ball around rc^T:

$$U_\alpha(r,c) = \{P \in U(r,c) : \mathrm{KL}(P \,\|\, rc^T) \le \alpha\}, \qquad d_{M,\alpha}(r,c) = \min_{P\in U_\alpha(r,c)} \langle P, M\rangle. \tag{3}$$

α	Radius of the ball, measured in nats of mutual information between X and Y.
U_α	Convex, because $\mathrm{KL}(\cdot \,\|\, rc^T)$ is convex in P (equivalently h is concave).

Economic and statistical meaning: α limits how informative the coupling may be. At $\alpha = 0$ the two bins are drawn independently and the cost is the average cost $r^T M c$ over all pairs. As α grows the plan may concentrate and the cost falls to the exact transport cost.

3.2 Limits of the entropic ball

PAPER Property 1 states that for α large enough, $d_{M,\alpha} = d_M$. The proof uses the bound $h(P) \geq \frac{1}{2}(h(r)+h(c))$, which gives KL $\leq \frac{1}{2}(h(r) + h(c))$ for all P, so $U_\alpha = U$ beyond that level.

INFERENCE The bound is correct and in fact weaker than necessary: since P determines both marginals, $h(P) \geq \max(h(r), h(c))$, so the ball already equals the polytope at $\alpha = \min(h(r), h(c))$. The conclusion is unchanged.

Property 2 states that for $\alpha = 0$ the set $U_0(r,c) = \{rc^T\}$ and $d_{M,0} = r^T Mc$. If M is a squared-distance matrix of a Euclidean configuration, $m_{ij} = \|\varphi_i - \varphi_j\|^2$, then

$$r^T Mc = \sum_{ij} r_i c_j \|\varphi_i - \varphi_j\|^2 = r^T u + c^T u - 2\, r^T K c, \qquad u_i = \|\varphi_i\|^2,\ K_{ij} = \langle \varphi_i, \varphi_j \rangle, \tag{4}$$

using $\sum_i r_i = \sum_j c_j = 1$. The first two terms are a function of r plus the same function of c, which is a negative definite kernel; the last is minus a positive definite kernel. Hence $r^T Mc$ is negative definite and, by Schoenberg's theorem, $e^{-t\, r^T Mc}$ is positive definite for all $t > 0$. This is the independence kernel.

CRITIQUE The statement of Property 2 needs M to be a matrix of squared Euclidean distances, whereas the paper's own text writes the proof with $\|\cdot\|^2$ in the first display and without the square in the second; the appendix also labels this proof as the proof of Property 1 and refers to U_1 where U_0 is meant. These are typographical, not substantive, but they show the appendix was not carefully proofread.

3.3 The metric theorem

PAPER Theorem 1 states that for every $\alpha \geq 0$ and every M in the metric cone, $d_{M,\alpha}$ is symmetric and satisfies the triangle inequality, and that $\mathbf{1}_{r \neq c}\, d_{M,\alpha}(r,c)$ satisfies all three distance axioms. The proof follows the classical gluing argument. Given optimal plans $P \in U_\alpha(x,y)$ and $Q \in U_\alpha(y,z)$, define the glued table

$$s_{ik} = \sum_{j:\, y_j > 0} \frac{p_{ij}\, q_{jk}}{y_j}. \tag{5}$$

Its margins are $\sum_k s_{ik} = \sum_j \frac{p_{ij}}{y_j} \sum_k q_{jk} = \sum_j p_{ij} = x_i$ and $\sum_i s_{ik} = \sum_j \frac{q_{jk}}{y_j} \sum_i p_{ij} = \sum_j q_{jk} = z_k$, so $S \in U(x,z)$. The entropic constraint is carried over by Lemma 1: the triple (X,Y,Z) with $\Pr(x,y,z) = \Pr(x,y)\Pr(y,z)/\Pr(y)$ is a Markov chain $X \to Y \to Z$, so the data processing inequality gives $I(X;Z) \leq I(X;Y) \leq \alpha$, hence $S \in U_\alpha(x,z)$. Then, using $m_{ik} \leq m_{ij} + m_{jk}$,

$$\begin{aligned} d_{M,\alpha}(x,z) \leq \langle S, M \rangle &= \sum_{ijk} m_{ik} \frac{p_{ij} q_{jk}}{y_j} \leq \sum_{ijk} (m_{ij} + m_{jk}) \frac{p_{ij} q_{jk}}{y_j} \\ &= \langle P, M \rangle + \langle Q, M \rangle = d_{M,\alpha}(x,y) + d_{M,\alpha}(y,z). \end{aligned} \tag{6}$$

The last equality uses the marginal identities $\sum_k q_{jk}/y_j = 1$ and $\sum_i p_{ij}/y_j = 1$.

CRITIQUE The argument is sound. Two remarks. First, only $I(X;Y) \le \alpha$ is used in the lemma; the constraint on Q is not needed, so the result holds with a weaker hypothesis. Second, the coincidence axiom fails: $d_{M,\alpha}(r,r) > 0$ for small α whenever $h(r) > 0$, because the plan must have mutual information at most α and so cannot be the diagonal plan. The paper handles this by multiplying by $\mathbf{1}_{r \neq c}$, which restores the axiom artificially and makes the function discontinuous at $r = c$. The printed chain of inequalities at the end of the proof of Lemma 1 writes the bound as "$\ge -\alpha$" in a form that does not follow line by line from the preceding one; the correct statement is the data processing inequality above.

3.4 The penalised problem and its solution

PAPER Since $d_{M,\alpha}$ has a hard constraint, the paper turns to the Lagrangian form with multiplier λ:

$$P^{\lambda} = \arg\min_{P \in U(r,c)} \langle P, M \rangle - \frac{1}{\lambda} h(P), \qquad d_M^{\lambda}(r,c) = \langle P^{\lambda}, M \rangle. \tag{7}$$

λ Inverse temperature. Large λ means little smoothing; $\lambda \to 0$ gives the independence table.

d_M^{λ} The transport cost of the smoothed plan. It is not the optimal value of the penalised objective, because the entropy term is excluded.

Derivation of the solution. Form the Lagrangian with multipliers $a, b \in \mathbb{R}^d$ for the two marginal constraints:

$$\mathcal{L}(P,a,b) = \sum_{ij} \left(\frac{1}{\lambda} p_{ij} \log p_{ij} + p_{ij} m_{ij} \right) + a^T (P\mathbf{1} - r) + b^T (P^T \mathbf{1} - c). \tag{8}$$

Setting $\partial \mathcal{L} / \partial p_{ij} = \frac{1}{\lambda}(\log p_{ij} + 1) + m_{ij} + a_i + b_j = 0$ gives

$$p_{ij}^{\lambda} = e^{-\frac{1}{2} - \lambda a_i} \, e^{-\lambda m_{ij}} \, e^{-\frac{1}{2} - \lambda b_j} = u_i \, e^{-\lambda m_{ij}} \, v_j, \qquad P^{\lambda} = \operatorname{diag}(u) \, e^{-\lambda M} \operatorname{diag}(v). \tag{9}$$

Positivity of p_{ij} is automatic, so the nonnegativity constraints are inactive (the paper writes the multipliers as nonnegative, although multipliers of equality constraints are unrestricted in sign; this does not affect the form). Strict convexity of $-h$ makes P^{λ} unique, and the theorem of Sinkhorn and Knopp (1967) guarantees that for the strictly positive matrix $K = e^{-\lambda M}$ there exist positive vectors u, v, unique up to a scalar, such that $\operatorname{diag}(u) K \operatorname{diag}(v)$ has marginals r and c. The vectors are found by alternating projections:

$$u \leftarrow \frac{r}{Kv}, \qquad v \leftarrow \frac{c}{K^T u}. \tag{10}$$

Algorithm 1 in the paper is a one-line rewriting of (10) in the variable $x = 1/u$, with $d_M^{\lambda} = \sum u \odot ((K \odot M) v)$. Each iteration costs two matrix-vector products, $O(d^2)$ operations, and a family of N target histograms $C = [c_1, \dots, c_N]$ is handled by replacing the vector v with a matrix, which turns the work into matrix-matrix products suited to graphics processors.

INFERENCE Two facts that the paper does not state but that frame its claims. First, the rate of convergence is linear and governed by the Birkhoff-Hopf contraction of the positive matrix K in Hilbert's projective metric; the contraction ratio tends to one as λ grows because K approaches a matrix with a wide dynamic range (Franklin and Lorenz, 1989). This predicts the observed rise in iteration count with λ (Figure 5 of the paper, and Table 2 below). Second, the gap to the exact distance obeys a simple bound. Let $P^\star$ be an optimal plan for (1). Optimality of P^λ for (7) gives $\langle P^\lambda, M\rangle - \frac{1}{\lambda} h(P^\lambda) \le \langle P^\star, M\rangle - \frac{1}{\lambda} h(P^\star)$, so

$$d_M(r,c) \le d_M^\lambda(r,c) \le d_M(r,c) + \frac{h(P^\lambda) - h(P^\star)}{\lambda} \le d_M(r,c) + \frac{h(r) + h(c)}{\lambda} \le d_M(r,c) + \frac{2\log d}{\lambda}. \tag{11}$$

The lower bound holds because P^λ is feasible for (1). The bound shows that the gap vanishes at rate $1/\lambda$ but also that it scales with $\log d$, which is why a fixed λ does not become a better approximation as the dimension grows.

3.5 From the hard constraint to the penalty

PAPER By Lagrangian duality each α corresponds to a $\lambda \in [0, \infty]$ such that $d_{M,\alpha}(r,c) = d_M^\lambda(r,c)$, and the entropy of P^λ decreases monotonically in λ, so the pair (r,c) can be matched by bisection. The paper then states that it uses only d_M^λ and calls it the Sinkhorn distance although it is not provably a distance.

CRITIQUE This is the central logical gap of the paper and is treated in Section 7. The correspondence between α and λ depends on the pair (r,c), so a fixed λ applied to all pairs corresponds to a different α for each pair, and Theorem 1, which needs the same α in $U_\alpha(x,y)$, $U_\alpha(y,z)$ and $U_\alpha(x,z)$, does not apply to the quantity actually computed.

4 Data and Empirical Strategy

4.1 Data

PAPER Three experiments are reported. The first uses the MNIST handwritten digits, reduced to 20×20 pixel grids and converted to histograms by dividing intensities by their sum, giving $d = 400$ bins. Subsets of $N \in \{3, 5, 12, 17, 25\} \times 10^3$ training digits are used. The second and third use synthetic histograms drawn uniformly from the d-simplex (Smith and Tromble, 2004), with random cost matrices obtained from d points of a spherical Gaussian in dimension $d/10$, rescaled by their median.

Table 1. Experimental design reported in the paper

Experiment	Data	Ground metric and parameters	Output
Classification	MNIST, $d = 400$, N from 3,000 to 25,000	Euclidean distance between pixel positions; $\lambda \in \{5, 7, 9, 11\}/q_{50}(M)$; 20 fixed iterations	Test error, 24 splits (4 folds, 6 repeats)
Gap to EMD	40^2 pairs of MNIST digits	As above	Relative gap $(d_M^\lambda - d_M)/d_M$
Timing	Uniform simplex draws, d up to the thousands	Random Gaussian cost, median-normalised; $\lambda \in \{1, 9\}$; tolerance 0.01	Seconds per distance; CPU against GPU
Iterations	Same as timing	Same	Iterations against λ

Source: reconstructed from Section 5 of the paper (official). The paper does not state the exact upper range of d in the timing study in the text, only that the Rubner et al. implementation cannot run beyond $d = 512$.

4.2 Empirical specification and identification

PAPER Classification uses support vector machines with the kernel $e^{-d/t}$ built from each distance, with t chosen on each training fold from the set $\{1, q_{10}(d), q_{20}(d), q_{50}(d)\}$, non-positive-definite kernel matrices repaired by adding a diagonal term, libsvm one-versus-one for the ten classes, and the cost constant chosen by two-fold cross-validation over $10^{-2:2:4}$. Comparators are Hellinger, χ^2, total variation, the squared-Euclidean Gaussian kernel, the exact EMD and the independence kernel. The identifying variation is purely predictive: all methods face the same splits, so differences in error are attributed to the distance. There is no causal claim, and the relevant question is whether the comparison is fair to every method (Section 6).

4.3 Data limitations and biases

- Single benchmark. MNIST has a natural spatial ground metric and strongly localised, roughly unimodal histograms; conclusions about superiority over the exact distance may not extend to histograms with distant modes or weak geometry.
- Tuning asymmetry. Sinkhorn has λ as an extra tuned parameter, chosen inside a four-value grid, while the exact distance has none; the kernel bandwidth t is common to all. Tuning budgets are therefore not identical.
- Positive definiteness. Neither the exact EMD kernel nor the Sinkhorn kernel is guaranteed positive definite, and the repair by diagonal shifts adds a method-dependent regularisation.
- Fixed 20 iterations. The classification results use a truncated iteration, so the object evaluated is an approximate P^λ, not the converged one; the implicit extra smoothing is not studied.
- Timing on synthetic simplex draws. Uniform draws are dense and unstructured, which favours matrix scaling; sparse histograms (common in practice) would help exact solvers and are not tested. Different languages (Matlab, mex/C) and an old CPU are mixed.

5 Results

5.1 Classification on MNIST

PAPER The Sinkhorn distance is reported to beat all other distances by a safe margin, the EMD included, with $\lambda = 9/q_{50}(M)$ most often selected. Figure 2 of the paper plots average test errors with shaded confidence intervals against N.

EVIDENCE The text gives no numerical error rates, and the figure was not visible in the rendering read, so magnitude, significance and economic (practical) relevance cannot be quantified from the material available. What is established by the design is the sign and rough size of the difference only to the extent the shaded intervals are separated, which cannot be checked here. The mechanism proposed is smoothness: an entropic plan spreads mass over nearby bins and so tolerates small deformations of a digit. That mechanism is plausible but is not isolated: the paper does not compare against a transport regularised by a different smoother, nor against EMD with a tuned thresholded ground distance, a standard remedy (Pele and Werman, 2009).

5.2 Gap to the exact distance

EVIDENCE With λ increasing, the relative excess $(d_M^\lambda - d_M)/d_M$ over 40^2 MNIST pairs falls but stays near ten percent even at large λ (Figure 3, boxplots). The paper concludes it neither expects nor desires closeness to the EMD at the preferred $\lambda = 9$. This is consistent with the bound in Equation (11), which implies a slowly vanishing gap of order $1/\lambda$.

5.3 Speed

EVIDENCE For large d on the same CPU core, Sinkhorn is reported as "more than 100.000 faster" than the EMD solvers at tolerance 0.01, and the GPU adds a further order of magnitude (Figure 4). The number is a ratio at the largest dimensions only, for random dense inputs, and compares a vectorised matrix routine against general-purpose solvers that compute an exact answer. It shows that matrix scaling is far cheaper per accuracy unit demanded here, not that exact transport is slow for all structured problems (for instance one-dimensional or tree metrics have closed-form or linear-time solutions).

5.4 Iterations against lambda

EVIDENCE The number of iterations needed for a change in x below 0.01 rises with λ (Figure 5). The authors recommend a fixed iteration count depending on λ for parallel platforms, since monitoring the change is expensive.

5.5 Independent numerical study by the reviewer

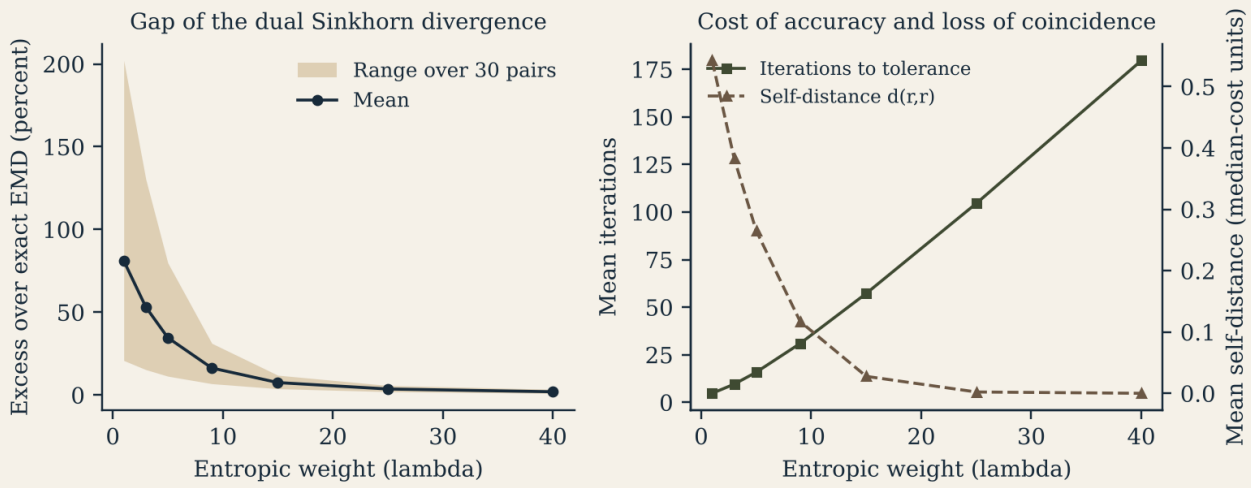
EXTENSION The paper's numbers cannot be reproduced from the text, so an independent small study was run to check the mechanisms. Setup (generated by the reviewer, not from the paper): $d = 64$ bins on an 8×8 grid, Euclidean ground metric divided by its median, 30 pairs of random smooth histograms (sums of three Gaussian bumps plus a small floor), exact EMD by linear programming (HiGHS), Sinkhorn iterated to a marginal error of 10^{-6} in the ℓ_1 norm, $\lambda \in \{1, 3, 5, 9, 15, 25, 40\}$ in median-cost units, which are the same units as the paper's $\lambda \cdot q_{50}(M)$ convention.

Table 2. **Reviewer's replication: gap to exact EMD, iterations and self-distance, by λ**

λ	Mean gap (percent)	Minimum	Maximum	Mean iterations	Mean $d_M^\lambda(r, r)$
1	81.0	20.8	202.0	4.7	0.542
3	53.2	15.3	130.3	9.5	0.383
5	34.6	11.4	79.8	15.8	0.265
9	16.3	6.8	31.2	30.9	0.117
15	7.5	3.8	11.9	57.2	0.028
25	3.6	2.0	5.7	104.5	0.0022
40	2.0	1.0	3.4	179.6	0.00005

Source: generated by the reviewer (Python, NumPy, SciPy), 30 random histogram pairs on an 8 by 8 grid; gap is $100\,(d_M^\lambda - d_M)/d_M$; the self-distance is in units of the median ground distance. Not data from the paper.

Figure 1. **Entropic gap, iteration count and self-distance as functions of lambda (8 by 8 grid, 30 histogram pairs)**



Source: generated by the reviewer; same simulation as Table 2. Left: relative excess of the dual Sinkhorn divergence over the exact EMD, with mean and range. Right: mean iterations to a marginal error of 10^{-6} and mean self-distance $d_M^\lambda(r, r)$ on the second axis.

Reading of Table 2 and the figure. The gap falls roughly as $1/\lambda$ (about 16 percent at $\lambda = 9$, 3.6 percent at $\lambda = 25$), as Equation (11) predicts, and in this smooth low-dimensional example it keeps falling instead of stalling at ten percent; the ten percent plateau in the paper therefore reflects either the MNIST geometry or the finite range of λ used, not an intrinsic floor. Iterations grow nearly linearly in λ (about 31 at $\lambda = 9$, 180 at $\lambda = 40$), so the speed advantage is bought with a trade between smoothing and iteration count. The self-distance $d_M^\lambda(r, r)$ is far from zero at $\lambda = 9$ (0.117 median-ground-distance units): the divergence does not vanish for identical histograms, confirming the failure of the

coincidence axiom for the computed quantity as well as for $d_{M,\alpha}$. A direct triangle-inequality search (150 random triples of smooth histograms and 400 random triples of sparse histograms with one to three atoms, for λ between 1 and 60) found no violation of $d_M^\lambda(x,z) \le d_M^\lambda(x,y) + d_M^\lambda(y,z)$. This does not prove the inequality; it only shows that violations, if any exist, are not easy to produce on this grid.

6 Robustness and Additional Tests

Table 3. **Robustness tests in the paper: concern, test, result and assessment**

Concern	Test reported	Result	Does it resolve the concern?
Sampling noise in the error comparison	4-fold cross-validation repeated 6 times, shaded intervals	Sinkhorn lowest across N	Partly. Intervals reflect split variability, not dataset variability; one dataset only.
Unfair tuning	Per-fold choice of kernel scale t, SVM cost and (for Sinkhorn) λ	Same grid logic for all distances	Partly. Sinkhorn has one more tuned parameter; the grids differ in size.
Choice of ground metric	Mahalanobis alternatives tried for the other distances; independence kernel with exponent $a \in \{0.01, 0.1, 1\}$	"None competitive" for Mahalanobis	Weak. Not reported numerically; ground metric for EMD and Sinkhorn is the same Euclidean grid, so the ground metric is not varied for the main contrast.
Sinkhorn merely approximates EMD	Gap boxplot against λ	About ten percent at large λ	Yes for the narrow point that the two differ; it does not show why the difference helps.
Speed advantage is an artefact of implementation	Comparison with two public EMD codes at several d and two λ	More than 10^5 at large d	Partly. Different languages and code maturity; random dense inputs only; no GPU baseline for exact methods.
Convergence of the iteration	Iterations to tolerance against λ	Count increases with λ	Yes for the observation; the linear rate is asserted by citation, not shown.

Source: assessment by the reviewer of the tests described in Section 5 of the paper.

The paper contains no placebo test, no alternative dataset, no alternative regulariser and no sensitivity of the classification result to the number of iterations, so robustness is limited. The most useful additional test would be to repeat the comparison on at least one dataset with a ground metric that is not spatial, and to include a tuned thresholded-ground-distance EMD.

7 Critical Evaluation

7.1 What the paper establishes

ESTABLISHED

Entropic regularisation turns the transport linear programme into a strictly convex problem whose solution has the diagonal scaling form (9) and is computed by iteration (10) in $O(d^2)$ per step. The hard-constrained distance $d_{M,\alpha}$ is symmetric and obeys the triangle inequality for every $\alpha \geq 0$ (proof checked above). The independence kernel is negative definite for squared-Euclidean M. Matrix scaling is vastly faster than general exact transport solvers on dense random inputs.

SUPPORTED BUT CONDITIONAL

Sinkhorn kernels classify MNIST digits better than the exact EMD. This depends on the tuning scheme, the spatial ground metric, the truncation to 20 iterations, and on one dataset.

SUGGESTIVE

Smoothness of the entropic plan is the reason for the better accuracy. The paper offers the maximum-entropy argument but no test that separates it from other explanations such as implicit bandwidth selection.

NOT ESTABLISHED

That the quantity actually computed, d_M^λ, is a distance or is positive definite as a kernel; that Sinkhorn is better than the exact distance in general; that no accuracy-speed trade-off exists ("we have not observed a need to establish a trade-off" is an absence of observation on one benchmark).

7.2 Strengths

STRENGTHS RECOGNISED BY THE AUTHORS

- Speed that is independent of the structure of the ground metric and vectorisable, which matters because it removes the main barrier to use.
- A principled interpretation through maximum entropy and mutual information, which links the method to statistics and information theory.
- Better performance on a standard benchmark.

STRENGTHS IDENTIFIED INDEPENDENTLY

- The identification of a closed form for the optimum, Equation (9), which reduces a constrained optimisation over d^2 variables to $2d$ scaling parameters; this is the reason the later literature built on it.
- Short, checkable proofs, and an honest remark that d_M^λ is not provably a distance.
- Four lines of Matlab suffice to implement the method, which gives high replicability and explains the adoption.

7.3 Weaknesses and critique

SINGLE MOST IMPORTANT WEAKNESS

Problem. The metric theorem is proved for $d_{M,\alpha}$ with one common α, but the experiments and Algorithm 1 compute d_M^λ with one common λ, and the paper itself concedes it is not proven to be a distance. **Why it matters.** Metric properties are the stated reason for calling the object a distance, for using it in nearest-neighbour or kernel methods, and for the claim that the regularisation is ”more robust”. **Potential bias.** The framing overstates what the evidence supports: the title and abstract speak of distances while the computed object fails coincidence (shown numerically above) and has no triangle-inequality guarantee; the exponentiated kernel may fail positive definiteness, which the paper silently repairs by adding a diagonal term. **How it could be tested.** Prove or disprove the triangle inequality for d_M^λ analytically, search for counterexamples with adversarial sparse histograms and non-metric M, and report the minimum eigenvalue of $e^{-d_M^\lambda/t}$ kernel matrices on the data used.

OTHER WEAKNESSES

- Experimental breadth. One dataset, one ground metric, no statistical test of differences between distances; without the numbers the margin cannot be assessed. Test: add datasets (text with embedding-based ground metrics, colour histograms) and paired tests across splits.
- Confounding of smoothing and bandwidth. The kernel $e^{-d/t}$ with tuned t may interact with the scale of d_M^λ, which is inflated relative to the EMD; a better score might come from rescaling and not from the geometry. Test: tune each distance over an identical rich grid of scales, or compare ranks of nearest neighbours directly.
- Bias of the quantity. The plan P^λ is a biased estimate of transport cost and is not the minimiser of anything linked to a statistical model; the entropy is not a feature of the data. Test: compare with debiased divergences that subtract self-terms.
- Numerical stability. $K = e^{-\lambda M}$ underflows for large λ or large M (the code adds no log-domain stabilisation), and null entries are handled only through selecting positive coordinates of r. Test: run in log-domain at large λ and compare.
- Linear convergence is asserted, not quantified; no iteration bound as a function of λ, d and $\|M\|_\infty$ is given.

8 Contradictions and Edge Cases

CRITIQUE Limit $\lambda \to 0$. Then $K \to \mathbf{1}\mathbf{1}^T$ and $P^\lambda \to rc^T$, so $d_M^\lambda \to r^T M c$. For $r = c = \delta_i$ this gives $m_{ii} = 0$, but for $r = c$ spread over many bins it gives a strictly positive value, the mean pairwise distance within the histogram. A distance that says two identical histograms are far apart in proportion to their spread is a failure of the intended geometry; the reviewer's Table 2 shows the same effect at moderate λ (0.117 median units at $\lambda = 9$, 0.542 at $\lambda = 1$).

CRITIQUE Limit $\lambda \to \infty$. The plan tends to the maximum-entropy vertex among optimal plans (Cominetti and San Martin, 1994), but K underflows and Algorithm 1 divides by zero in floating point long before the limit is reached; the iteration count rises (Table 2) so the cheap regime and the accurate regime are mutually exclusive. A quick calculation shows the practical tension: with $d = 400$ as in MNIST, Equation (11) gives an additive gap of up to $2 \log 400/\lambda \approx 12/\lambda$ in ground-metric units, so a guaranteed gap of one tenth of a median pixel distance needs $\lambda \approx 120$, at which entries $e^{-\lambda m}$ for distant bins are of order e^{-240} or smaller for m above two median units, uncomfortably close to the floating-point range of double precision.

CRITIQUE Dimension scaling. The bias term is proportional to $\log d$, so for a fixed λ the smoothing becomes relatively more important as bins are refined: refining a grid by a factor of two in each direction multiplies d by four while leaving the ground distance in continuous units unchanged. The classification advantage found at $\lambda = 9$ on a 20×20 grid therefore need not carry over to 40×40 images with the same λ in physical units.

CRITIQUE Disjoint supports. For two histograms with disjoint supports far apart, the EMD is the distance between the supports, but P^λ for moderate λ spreads mass to all pairs and d_M^λ is pulled toward $r^T M c$, compressing large distances. The distance saturates for dissimilar histograms, which may help robustness but erases information about how dissimilar they are.

CRITIQUE Alternative mechanisms. The accuracy gain could equally come from the implicit conversion of a hard assignment problem into a kernel-density-type weighting, or from the 20-iteration truncation which acts as early stopping and regularises further. None of these is excluded by the experiments.

9 Further Literature

The table lists the works that precede, compete with, correct or extend the paper, each with the specific idea it shares with it.

Table 4. **Literature connections**

Paper	Shared idea	Relationship	Why it matters
Sinkhorn and Knopp (1967)	Scaling of a positive matrix to given marginals	Foundation of the algorithm	Gives existence, uniqueness and convergence of u, v.
Wilson (1969), Erlander and Stewart (1990)	Entropy-maximising trip distribution (gravity model)	Preceded: same solution form $u_i e^{-\lambda m_{ij}} v_j$	Shows the method is a rediscovery in machine learning.
Rubner, Tomasi and Guibas (2000)	Earth mover's distance for image retrieval	Established the exact comparator	Defines the baseline and its cost.
Pele and Werman (2009)	Fast EMD with thresholded ground distance	Competing speed-up that keeps exactness	Tests whether restrictions alone suffice.
Ferradans et al. (2014)	Regularised discrete transport	Concurrent alternative regulariser	The only direct comparison of smoothing choices.
Cominetti and San Martin (1994)	Exponential penalty trajectory in linear programming	Theory of the limit $\lambda \to \infty$	Characterises the limit plan and rate.
Cuturi and Doucet (2014)	Barycentres of histograms using Sinkhorn	Applied and extended	Extends from pairs to averages.
Benamou et al. (2015)	Iterative Bregman projections	Reinterpreted the iteration	Explains Sinkhorn as KL projections; handles barycentres.
Altschuler, Weed and Rigollet (2017)	Near-linear time approximation of OT	Quantified convergence	Gives iteration bounds absent from the paper.
Genevay, Cuturi, Peyre and Bach (2016)	Stochastic optimisation of regularised OT	Extended to continuous measures	Removes the dependence on histograms.
Feydy et al. (2019)	Sinkhorn divergence with self-terms subtracted	Corrected the bias and coincidence failure	Directly answers the weaknesses noted above.
Peyre and Cuturi (2019)	Monograph on computational OT	Consolidated the field	Gives the modern treatment of every topic here.
Esfahani and Kuhn (2018)	Distributionally robust optimisation over a Wasserstein ball	Applied different use of transport distance	Link to finance and risk, for the exact distance.

Source: related works are cited from the reviewer's prior knowledge of the literature and were not re-opened in this session; only the reviewed paper was read in full. Bibliographic details should be verified before citing.

10 Research Gaps

10.1 Paper-level gaps

- **Gap:** no proof or counterexample for the metric properties of d_M^λ. *Why it matters:* it decides whether the object is usable in metric-space methods. *Existing limitation:* the pair-dependent correspondence between α and λ. *Way forward:* bound the violation of the triangle inequality in terms of λ and $\log d$, or use the debiased divergence.
- **Gap:** no numerical report of the error rates, and no uncertainty on the gap between methods. *Why it matters:* claims of superiority cannot be assessed. *Existing limitation:* figures only. *Way forward:* replicate on MNIST and report tables with paired tests.

10.2 Literature-level gaps

- **Gap:** how to choose λ from data, rather than by cross-validation. *Why it matters:* the choice trades bias against speed and accuracy. *Existing limitation:* no theory for the statistical effect of smoothing. *Way forward:* study the estimation error of the regularised cost between empirical measures as a function of λ and sample size.
- **Gap:** ground metric learning jointly with the downstream task. *Why it matters:* the advantage over bin-wise distances rests on M. *Existing limitation:* M is fixed by hand. *Way forward:* differentiate through the scaling iterations.

10.3 Methodological gaps

- **Gap:** numerically stable computation at large λ. *Why it matters:* underflow limits accuracy. *Existing limitation:* the direct kernel $e^{-\lambda M}$. *Way forward:* log-domain updates with stabilisation, or scaling schemes.
- **Gap:** structured histograms and sparse supports. *Why it matters:* real data is often sparse, where the cost comparison with exact solvers changes. *Existing limitation:* only dense random inputs. *Way forward:* benchmark on sparse and low-dimensional geometry.

11 Research Opportunities

Five projects follow from the weaknesses and limits identified above. Each is stated with its hypothesis, data, method and main risk, and they are ranked at the end of the section.

Idea 1. Order book shape distance for short-horizon price prediction

RESEARCH QUESTION. Does the Sinkhorn divergence between limit order book depth profiles, with tick distance as ground metric, predict the next mid-price move better than bin-wise distances or imbalance measures?

MOTIVATION. Book snapshots are histograms over price levels where neighbouring levels are close substitutes, which is the structure a ground metric handles.

HYPOTHESIS. Nearest-neighbour or kernel forecasts built on transport geometry have higher out-of-sample directional accuracy.

ECONOMIC MECHANISM. Queue depletion at nearby price levels has similar price impact, so shifting depth by one tick should change the distance little.

LITERATURE CONNECTION. Cont, Kukanov and Stoikov (2014) on order flow imbalance; Cuturi (2013).

DATA. Level-2 or level-3 book data for liquid futures or equities, for example a vendor sample over a year.

VARIABLES. Normalised depth by tick on each side, next-interval mid-price change, spread, volume.

METHOD. Kernel regression or support vector classification with $e^{-d_M^{\lambda}/t}$, compared with Euclidean, Hellinger and imbalance baselines, walk-forward.

IDENTIFICATION STRATEGY. Predictive; strict time-ordered splits, purged cross-validation, multiple-testing control.

EXPECTED CONTRIBUTION. A tested use of transport geometry in microstructure.

MAIN RISK. Predictability may be tiny after costs and the book may be dominated by imbalance alone.

Idea 2. Correlation-aware distance between portfolios

RESEARCH QUESTION. Can a transport distance between weight vectors, with ground metric $m_{ij} = \sqrt{2(1-\rho_{ij})}$, measure turnover and benchmark deviation better than the ℓ_1 distance?

MOTIVATION. Selling one stock and buying a close substitute is cheap in risk terms but costs full ℓ_1 turnover.

HYPOTHESIS. The transport distance between consecutive portfolios predicts tracking error changes better than ℓ_1 turnover.

ECONOMIC MECHANISM. Substitutability within correlated clusters.

LITERATURE CONNECTION. Mantegna (1999) distance; Markowitz (1952); Cuturi (2013).

DATA. Daily returns for an equity index universe, long-only fund holdings.

VARIABLES. Weights, rolling correlations, tracking error, realised turnover cost.

METHOD. Compute d_M^{λ} between portfolios (positive weights, normalised), regress tracking error changes on it and on ℓ_1 turnover.

IDENTIFICATION STRATEGY. Panel regressions with fund fixed effects; placebo with randomised correlation matrices.

EXPECTED CONTRIBUTION. A new transaction-risk-aware distance for portfolio management.

MAIN RISK. Correlation instability and the restriction to nonnegative weights.

Idea 3. Entropic ball as an ambiguity set for robust portfolio choice

RESEARCH QUESTION. Does replacing the Wasserstein ball by an entropic transport ball give a tractable robust problem with comparable out-of-sample performance?

MOTIVATION. Wasserstein robust optimisation is accurate but computationally heavy.

HYPOTHESIS. The smoothed ball gives less conservative portfolios and lower variance of the out-of-sample Sharpe ratio.

ECONOMIC MECHANISM. Investors guard against estimation error in the return distribution.

LITERATURE CONNECTION. Esfahani and Kuhn (2018); Rockafellar and Uryasev (2000); Artzner et al. (1999).

DATA. Daily returns for a few hundred stocks; Monte Carlo with known truth.

VARIABLES. Radius of the ball, λ, conditional value at risk, turnover.

METHOD. Dual reformulation with the log-sum-exp structure of entropic transport; compare with the empirical and Wasserstein solutions.

IDENTIFICATION STRATEGY. Rolling out-of-sample comparison and simulation with known distribution.

EXPECTED CONTRIBUTION. Faster robust portfolio construction.

MAIN RISK. The duality may not give a closed form, and the gain over simple shrinkage may be small.

Idea 4. Triangle-inequality violation bound for the dual divergence

RESEARCH QUESTION. What is the largest possible violation of the triangle inequality by d_M^λ as a function of λ, d and M?

MOTIVATION. Closes the central gap in the paper.

HYPOTHESIS. The violation is bounded by a term of order $\log d/\lambda$ and vanishes as $\lambda \to \infty$.

ECONOMIC MECHANISM. Not applicable; a mathematical question.

LITERATURE CONNECTION. Cuturi (2013) Theorem 1; Feydy et al. (2019); Equation (11).

DATA. Synthetic: random and adversarial histograms, structured metrics.

VARIABLES. Maximum ratio of left to right side of the inequality.

METHOD. Analytical bound via (11) and a global search for counterexamples.

IDENTIFICATION STRATEGY. Proof or explicit counterexample.

EXPECTED CONTRIBUTION. A precise statement of what kind of distance is computed.

MAIN RISK. The bound may be loose, and the reviewer search in Section 5 found no violation.

Idea 5. Distribution distance between cross-sectional return states

RESEARCH QUESTION. Does the Sinkhorn distance between the cross-sectional distributions of factor exposures or returns on consecutive days signal regime change earlier than volatility measures?

MOTIVATION. Regime detection usually uses summary statistics; a transport distance uses the entire distribution with a geometry on characteristics.

HYPOTHESIS. Spikes in distance precede drawdowns in factor portfolios.

ECONOMIC MECHANISM. Crowding and de-risking shift mass across the characteristic space.

LITERATURE CONNECTION. Fama and French (1993); Cuturi (2013).

DATA. CRSP or a free substitute, monthly or daily returns binned by size and book-to-market.

VARIABLES. Binned return or weight histograms, distance series, factor returns, volatility.

METHOD. Time series of d_M^{λ} between adjacent states; predictive regressions and event study around drawdowns.

IDENTIFICATION STRATEGY. Predictive with out-of-sample tests, controlling for volatility and turnover.

EXPECTED CONTRIBUTION. A geometry-aware state variable for asset pricing and risk management.

MAIN RISK. Binning choices drive results, and look-ahead in bin construction.

11.1 Ranking

Table 5. **Ranking of research ideas (scores from 1 to 5, assigned by the reviewer)**

Idea	Novelty	Feasibility	Data	Method	Importance	Contribution	Total
1. Order book shape distance	4	3	3	4	4	4	22
2. Correlation-aware portfolio distance	4	5	5	4	4	4	26
3. Entropic ambiguity set	4	3	4	3	4	4	22
4. Triangle violation bound	3	4	5	5	3	3	23
5. Cross-sectional state distance	3	4	4	3	3	3	20

Source: reviewer's subjective scoring.

The top three are Idea 2, Idea 4 and Idea 1 (with ties at 22 broken in favour of the idea with the more concrete test). Idea 2 is first because data are free and abundant, the economic question is clear, and the contrast with the ℓ_1 benchmark is sharp. Idea 4 is second because it is cheap, rigorous and closes a stated gap. Idea 1 is third because it connects to a large literature and has a real prediction target, though data access costs and weak signals reduce feasibility.

12 Learning Framework

12.1 Prerequisite concepts

- Linear programming duality and the structure of the transportation polytope.
- Shannon entropy, Kullback-Leibler divergence, mutual information and the data processing inequality.
- Lagrange multipliers and strictly convex optimisation.
- Negative definite and positive definite kernels, and Schoenberg's theorem.
- Metric axioms and the gluing lemma.

12.2 Core concepts learned

- Entropic smoothing converts a combinatorial problem into a diagonal scaling problem.
- A hard constraint and a penalty are linked by duality, but the multiplier depends on the data pair.
- A computed quantity may lose the properties proved for the idealised one.
- Speed comes from the algebra of matrix products, not from approximating the objective.

12.3 Important equations

Equation (2) ties entropy to mutual information; Equation (9) is the key structural result; Equation (10) is the algorithm; Equation (6) is the metric proof; Equation (11) is the reviewer's bound on the bias.

12.4 Methods learned

Maximum-entropy regularisation of linear programmes; Sinkhorn-Knopp iteration; gluing arguments for metric properties; kernel construction from non-Euclidean distances and repair of indefinite kernels; cross-validated SVM comparisons across distances.

12.5 Questions to test understanding

COMPREHENSION

1. Why does the exact transport optimum lie at a vertex with at most $2d - 1$ nonzero entries?
2. What is $U_0(r, c)$ and what is the distance for $\alpha = 0$?
3. What are the two operations in one Sinkhorn iteration and what is their cost?

ANALYSIS

1. Derive the form of P^λ from the Lagrangian and explain why the nonnegativity constraints are inactive.
2. Why does the proof of Theorem 1 fail for d_M^λ with a fixed λ?
3. Derive the bound (11) and discuss when it is tight.

RESEARCH

1. Design an experiment that separates the effect of entropy from the effect of any smoothing on classification accuracy.
2. How would you learn the ground metric M jointly with a classifier?
3. What would a debiased version of d_M^λ look like, and which of the paper's properties would it restore?

12.6 Papers to read next

Benamou et al. (2015); Altschuler, Weed and Rigollet (2017); Feydy et al. (2019); Peyre and Cuturi (2019).

13 Conclusion

13.1 One-sentence thesis

Entropic regularisation of the transport problem yields a quantity computable by matrix scaling several orders of magnitude faster than exact solvers, which the author argues is a better histogram distance for learning, though its metric properties are proved only for a related, different object.

13.2 Three most important insights

- The optimum of the smoothed problem has the form $\mathrm{diag}(u)e^{-\lambda M}\mathrm{diag}(v)$, which makes the computation a matter of two matrix-vector products per step.
- The price of smoothing is a bias of order $\log d/\lambda$ and a failure of coincidence, visible in the reviewer's simulations.
- Metric proofs for the constrained form do not transfer to the penalised form that is computed.

13.3 Strongest evidence

The closed-form scaling solution and the speed comparison, which follow from algebra and timing and do not depend on tuning.

13.4 Biggest weakness

The computed divergence is not proved to be a distance, while the framing and the title invite the reader to treat it as one.

13.5 Most important research gap

A theory of the statistical and geometric effect of the smoothing parameter: how λ should be chosen, what bias it introduces, and which properties survive.

13.6 Best research opportunity

Idea 2: a correlation-aware transport distance between portfolios, which is cheap to test with public data and has a clear benchmark.

13.7 Three papers to read next

Benamou et al. (2015); Feydy et al. (2019); Altschuler, Weed and Rigollet (2017).

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